

Assignment 8

Problem Statement: Develop a text-to-speech model using Tacotron or WaveNet.

Introduction

Text-to-Speech (TTS) is a technology that converts written text into spoken audio using artificial intelligence. It plays a crucial role in applications such as virtual assistants, accessibility tools, and automated customer service systems. Modern TTS systems use deep learning models to generate natural and human-like speech.

In this assignment, a text-to-speech system is developed using advanced models like Tacotron or WaveNet. Tacotron focuses on converting text into spectrograms, while WaveNet generates high-quality audio waveforms. By combining these approaches, the system produces clear, natural, and expressive speech from textual input.

Objective

- To understand the concept of text-to-speech systems.
- To implement a TTS model using Tacotron or WaveNet.
- To convert textual input into natural-sounding speech.
- To analyze the quality of generated audio.
- To explore applications of speech synthesis in real-world systems.

Theory

1. Text-to-Speech (TTS)

Text-to-Speech systems transform written text into audible speech. The process involves linguistic analysis and audio waveform generation.

Basic Steps:

1. Text preprocessing (tokenization, normalization)
2. Conversion to phonemes
3. Acoustic modeling
4. Waveform generation

2. Tacotron

Tacotron is an end-to-end neural network model for TTS.

Working:

- Converts text into a **mel-spectrogram**
- Uses sequence-to-sequence architecture with attention mechanism

Advantages:

- Produces natural-sounding speech

- End-to-end learning

Limitations:

- Requires a vocoder (e.g., WaveNet) to generate audio

3. WaveNet

WaveNet is a deep generative model for producing raw audio waveforms.

Working:

- Generates audio sample-by-sample
- Uses dilated causal convolutions

Advantages:

- High-quality and realistic speech
- Captures fine audio details

Limitations:

- Computationally expensive
- Slower generation

4. Combined TTS Pipeline

A modern TTS system often combines both models:

1. **Input Text**
2. **Tacotron:** Converts text → spectrogram
3. **WaveNet (Vocoder):** Converts spectrogram → audio waveform
4. **Output:** Natural speech

5. Evaluation Metrics

- **MOS (Mean Opinion Score):** Human evaluation of audio quality
- **WER (Word Error Rate):** Measures intelligibility
- **Naturalness:** Human-like speech quality
- **Clarity:** Pronunciation accuracy

6. Advantages

- Produces human-like speech
- Supports multiple languages and voices
- Useful for accessibility applications

7. Limitations

- High computational requirements

- Needs large datasets for training
- May produce unnatural speech if not properly trained

8. Applications

- Virtual assistants (e.g., voice assistants)
- Audiobooks and narration systems
- Assistive technology for visually impaired users
- Customer service automation
- Language learning tools

Conclusion

Developing a text-to-speech system using Tacotron and WaveNet demonstrates the capabilities of deep learning in speech synthesis. Tacotron efficiently converts text into intermediate representations, while WaveNet generates high-quality audio, resulting in natural and intelligible speech.

Although these models require significant computational resources and data, they provide state-of-the-art performance in TTS systems. This assignment highlights the importance of combining different deep learning techniques to build powerful and practical AI applications.